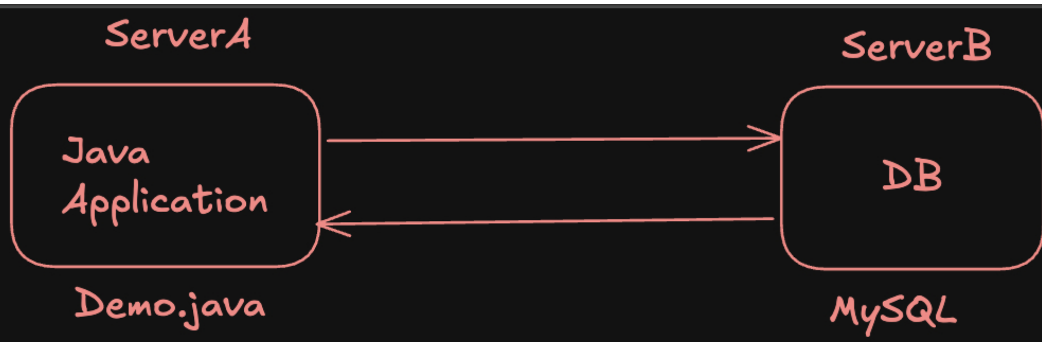


JVM

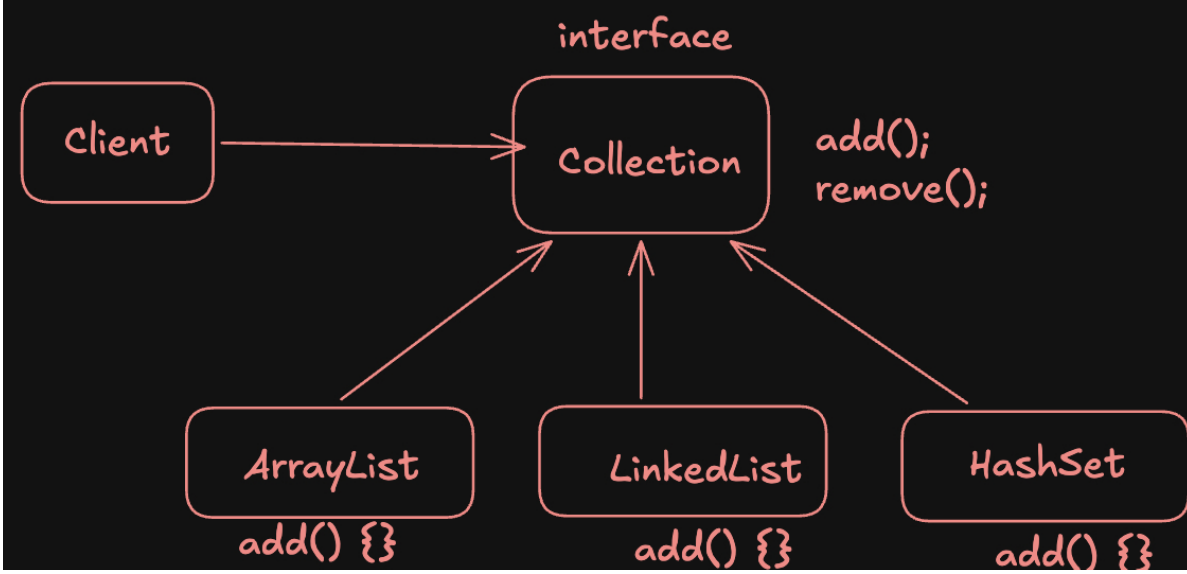
Select * from student;

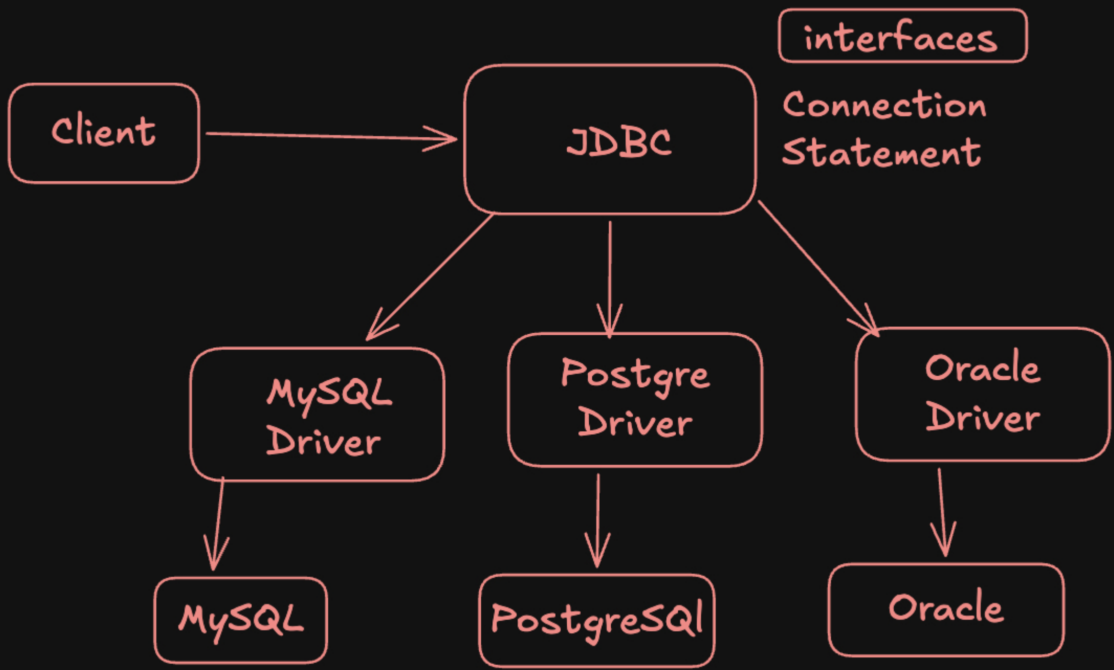
MySQL :
class and Interfaces

PostgreSQL
Class and Interfaces

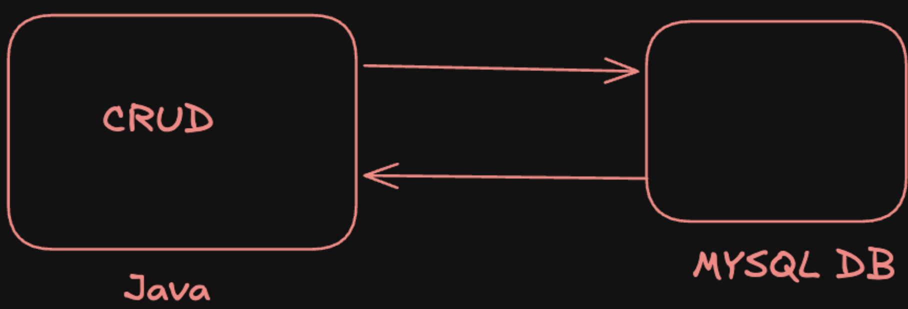
JDBC : Java Database Connectivity

import java.sql





- How a connection is established
- How authentication information is sent
- How SQL queries are packaged
- How different SQL data types are represented
- How query results are returned
- How errors are communicated
- How transactions are controlled



MySQL Server
MySQL CLient (DBEaver)

Connection ← Statement

Java App → DB

Sttring sql = "Select * from student";

Statement

↓

executeUpdate() --> CREATE, UPDATE, DELETE
executeQuery() --> READ (SELECT)
execute() --> General purpose (CRUD)

ResultSet

id	name	email	age
1	Aditya	a@gmail.com	28
2	Rohit	r@gmail.com	26
3	Aman.	a.@gmail.com	28

resultSet.getLong("id")
resultSeet.getString6("name")

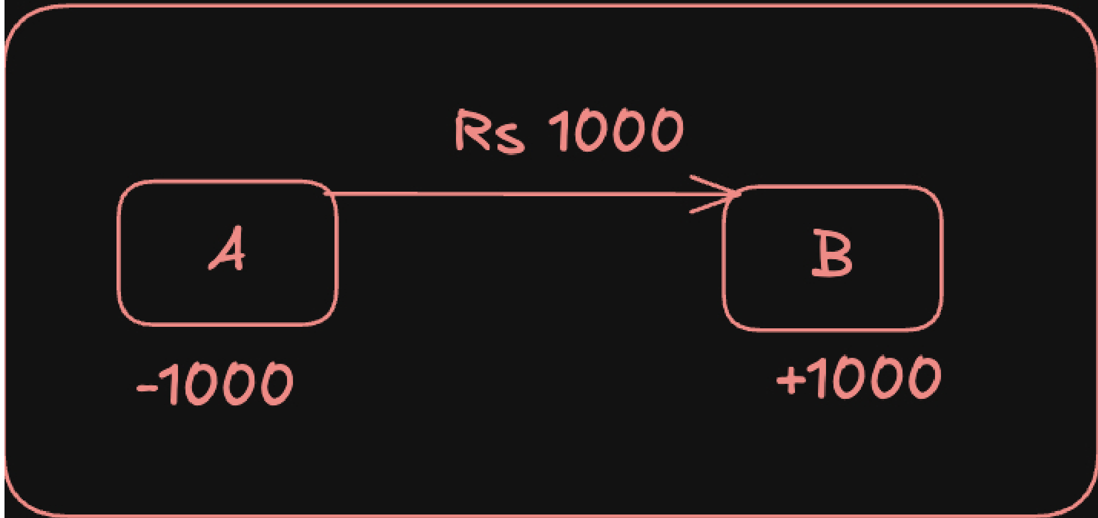
resultSet.getLong(1)

resultSet.next() --> True, False

statement.execute(sql)

→ True (ResultSet --> SELECT)

→ False (Int --> Row affected)




```
connection.setAutoCommit(false);

try {
    // first
    // second
    connection.commit()
}
catch() {
    connection.rollback();
}
```